

A gas heater with a black metal frame and a mesh screen, sitting on a white table outdoors. The heater is lit, with orange flames visible through the mesh. The background shows a green lawn and a blue fence.

TECHNICAL OCCUPATIONAL MECHANICAL TECHNOLOGY WELDING

Learner's Guide

Year 3 / Term 1

INTRODUCTION TO MECHANICAL TECHNOLOGY: WELDING

2.1 What is Welding?

Welding is a fabrication process that joins materials, usually metals and other materials. This is often done by melting the work pieces and adding a filler material to form a pool of molten material that cools to become a strong join.

Metalworking is the process of working with metals to create individual parts, assemblies, or large scale structures.

2.2 Topics to be studied in Welding

1. Safety and first aid – Occupational Health and Safety (OHS) Act;
2. Workshop environment
3. Drawings (rough sketches, simple drawing ISO, Ortho and CAD welding related)
4. Tools and equipment (hand, power and machine)
5. Materials – ferrous and non-ferrous materials
6. Joining methods – permanent and semi-permanent.
7. Welding (operations/procedures and equipment).

2.3 Specific Aims:

The learner is able to:

1. Adhere to and identify safe working practices and demonstrate safe working conditions daily, also adopting proper safety and first aid procedures.
2. Demonstrate knowledge of the welding industry and its productivity requirements, by applying appropriate work-procedures.
3. Understand and interpreting work instructions and drawings for the completion of projects.
4. Apply welding technology, techniques, processes and skills, as applied in the fabrication and welding industry, using appropriate tools and measuring equipment.
5. Identify Ferrous & Non-Ferrous metals the differences, applications and uses of them.

6. Demonstrate the different welding processes: Arc (AC/DC), MIG, TIG, Brazing, Soldering and spot welding.

Revision of Safety, first aid HIV and AIDs (OHS Act)

Demonstrate understanding of the following:-

What Occupational Health and Safety requirements do companies have to comply with in South Africa?

In South Africa, companies are required to comply with the [Occupational Health and Safety Act, 85 \(OHSA\) of 1993](#). This act establishes the rights and duties of both employers and employees with regard to health and safety in the workplace.

Under the OHS Act, employers are required to provide a safe and healthy working environment for their employees. This includes taking measures to prevent accidents and injuries, as well as providing necessary protective equipment and ensuring that all machinery and equipment is in good working order. Employers are also responsible for providing training and information to their employees about potential hazards and how to avoid them.



SAFETY PROCEDURES

Safe Operating / Safe work Procedures should be put together for workplaces. This aids workplaces to function in an organized fashion, these are documented safety guidelines for procedures and tasks to ensure that activities in the workplace are conducted safely. All activities must abide to these guidelines. Without these protocols, incidents and injuries would be the order of the day.

BEFORE SAFE OPERATING PROCEDURES CAN BE IMPLEMENTED, THE FOLLOWING STEPS HAVE TO BE TAKEN:

Identify the hazards and the risk associated with them - Risk Assessment

Evaluate the hazards involved when performing a task

Establish what personal protective equipment is required,

Document the steps required so that an activity is completed without any incidents.

The importance of Safe Operating Procedures is that they provide the tools for teaching employees how to work consistently. These Safe Operating Procedures are for sample purposes to be used when you are developing procedures specific to your organization. They also educate employees on what, when and how to perform their duties without risking their lives and that of others.

Examples of Safe Operating Procedures

Regular updating of work policies and instructions

Ensuring that instructions and procedures are clear and well understood

Empower employees to report any potential hazards, hazards, near misses they come across

Ensure induction training is provided to all new employees in the workplace

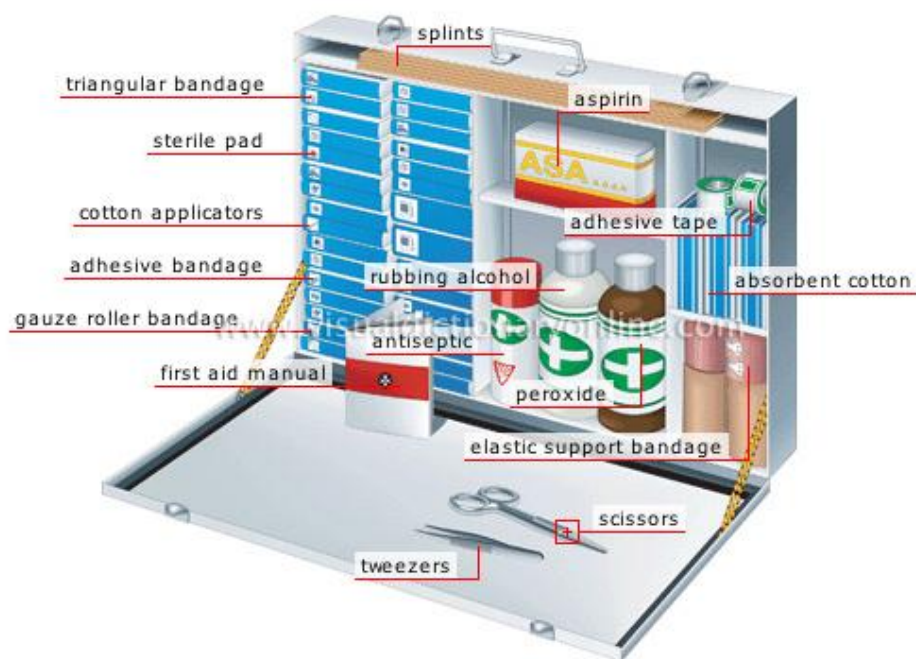
Provide regular refresher training for all employees

The regular conducting of fire drills and evacuations to establish if employees are well informed of steps to take

Ensure all employees are aware of (the location of) vital services in the workplace e.g. fire siren, fire equipment, first aid kit, emergency exit.

https://youtu.be/Ek4RI_zZA8I

FIRST AID BOX



Personal Protective clothes

Work-wear Clothing

Personal protective work-wear comes under the umbrella of PPE (personal protective equipment) and is designed to ensure the safety of you and others using either protective materials or disposable units of clothing or accessories, depending on the application.

PPE / PPC is important as it will protect you from:

- Potential flying objects
- Protection from welding lights
- Eye infections, skin infections
- Burns
- Physical elements

Welding apron 	Weld red leather gloves 	Welding goggles 	welding goggles with square flip front lens 
Welding leg guard 	welding knee pads 	welding screen 	leather ankle spat 
Red frame-brazing gas Cutting welding goggles 	Adjustable welding helmet 	weld blue leather gloves 	weld prime leather gloves 
auto darkening true colour welding helmet 	green lined welders gloves 	Welding jacket 	flip front welding helmet 

 leather knee spat	 Weld spectacles	 welders hood	 skull cap
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COLOURS THAT SIGNS ARE MADE OUT OF AND MEANING THERE OF.

There are four colours associated with health and safety signage: red, yellow, blue and green.

The reader can easily interpret the importance of the sign from the colour displayed which in turn influences their reaction.

What colours do health and safety signs have to be?

A health and safety sign is any sign that is used to convey information about the safety of a given area or item. Some of the most recognizable examples include:

- Hard hat area signage
- Wet floor signage
- Suspended loads signs
- The COSHH symbols and signage

They are used to convey the danger of a certain situation or sometimes to provide a reminder for when workers need to wear the correct PPE. The different types of health and safety signs all have their different designs and colours which are used to help differentiate them from one another.

Why are safety sign colours important?

The colour of health and safety signs is a very important aspect of what their meaning holds. Some colours signify rules whilst others signify danger.

Mandatory safety signs (Blue)	Mandatory health and safety signs signal the need for certain behaviours. It conveys information that you must comply with to be safe. These signs must be blue and usually have a white symbol on a blue background. They are often used to notify those of the need to use PPE.	
Fire safety signs (Red)	Fire safety signs provide emergency information on the escape route needed in the event of a fire. These health and safety signs can also provide information on areas that may be susceptible to fires. The law that requires that these health and safety signs are red.	
Prohibition safety signs (Red)	Prohibition signs are used to signify the prohibition of a certain act or event. These health and safety signs are required to be red. Prohibition signs show workers what they cannot do in a workplace and usually have a black safety symbol within a red crossed circle.	

Emergency escape/first aid safety sign (Green)	Emergency escape and first aid signs are used to show information that indicates a safe escape route or the route to first aid facilities. These health and safety signs are required to be green, and usually have a white symbol on a green background.	
Warning: Safety signs (Yellow)	These health and safety signs are used to show a specific danger that may be present within a workplace. They can indicate a general hazard/danger and are sometimes known as hazard/caution signs. These health and safety signs need to be yellow/amber and are usually characterised by a black symbol on a yellow background.	

Health and Safety Signs & Their Meanings

Red Signs

When the colour red is used in a safety sign it is usually a prohibitory sign which aims to draw the viewers' attention to a dangerous situation nearby or one that may arise through their actions. Prohibitory signs are usually round in shape and feature a thick red edge with a diagonal line through a black and white picture, depicting the prohibited behaviour.



Yellow or Amber Signs

As with traffic lights, a yellow or amber safety sign depicts a warning, which alerts the viewer to take precaution or care in their environment. Commonly seen when using hazardous chemicals, a warning sign is triangular in shape.

In the work place, it is the school's responsibility to ensure all staff and learners understands the meaning of all safety signage displayed within their facilities even if it the sign's message appears to be self-explanatory.



Blue Signs

Blue safety signs are known as mandatory signs, which demonstrate that a specific behaviour must be undertaken before proceeding, this might be applying safety glasses when working with chemicals or wearing a hard helmet when entering a construction site.

Mandatory safety signs are circular in shape and are made up of over 50% blue background with a white pictogram displaying the instruction.



Green Signs

Green safety signs provide information to the reader which highlight safety, showing exit routes, first aid areas or stations. These signs are rectangular in shape with white imagery, clearly showing the direction the person needs to move in or the safety facilities provided at a location.



Textual prompts can be used in conjunction with the images displayed on any of these signs to ensure the correct message is delivered to the viewer

General safety practices and rules that govern the Welding workshop on a daily basis.

What are Welding Safety Rules?

The Occupational Safety and Health Administration (OSHA) outlined specific requirements for welding, cutting, and brazing. Consider starting to apply these rules below in daily business operations.

1. Conduct in a designated safe location

A welder should observe their surroundings or working environment prior to welding. It is best to observe whether any flammable equipment or materials are present within the vicinity of the welding area. The considered safe distance from the welding area is 10 meters. If the object to be welded or cut cannot readily be moved, a work around OSHA recommended is to transfer movable fire hazards away from

the welding area.

2. Protect yourself from fire hazards

If all the fire hazards cannot be removed, then appropriate safeguards should be in place to protect the immovable fire hazards and to confine heat, sparks, and hot slags that are produced by welding.

3. Consider the risks

There are various risks associated with welding, (e.,g exposure to fumes and UV radiation). Welding risks may vary depending on the area where it will be done, so it is recommended that a risk assessment be carried out prior to work. This will help in setting up controls to manage or eliminate the risks. Otherwise, no welding and cutting work should be permitted.

4. Maintain your equipment

A welder or helper working on platforms, scaffolds, or runways shall be protected against falling by the use of railings, safety belts, life lines, or some other equally effective safeguards. Welders shall place welding cable and other equipment so that it is clear of passageways, ladders, and stairways.

5. Protect yourself with the correct PPE

Welders should always wear the appropriate PPE to safeguard against hazards that can cause physical injury or illness. Some example of PPE's that welders should wear are:

- Long-sleeved shirts
- Long pants with no cuffs
- Welding helmet or goggles
- Respirators
- Ear muffs/plugs
- Boots & gloves

6. Check for ventilation

Another means of preventing welding incidents and protecting learners or employees is proper ventilation. Ventilation is mainly used for removing air contaminants from a worker's work area, preventing accumulation of flammable or combustible gases or vapors, and preventing oxygen rich or deficient atmospheres.

7. Protect yourself from fumes and gases

Local exhaust or general ventilating systems shall be provided and arranged to keep the amount of toxic fumes, gases, or dusts below the maximum allowable concentration. If necessary, welders should wear a respirator to protect themselves from breathing in harmful substances.

8. Protect other workers

After welding operations are completed, the welder shall mark the hot metal or provide some other means of warning other workers.

9. Read warning labels, relevant documents and keep training

Ensure that each welder has access to labels on containers of such materials and safety data sheets, and is trained when new equipment and techniques are adopted.

10. Enforce Safety Procedures

Hold yourself and those around you accountable when it comes to following safety guidelines. Report on any safety violation activities in the best interest of you and those you share a workspace with.

Welding Safety Checklist

What is a Welding Safety Checklist?

A welding safety checklist is a tool used to assess workers' safety throughout the entire welding, cutting, and brazing processes. It helps ensure compliance with regulatory standards including wearing of appropriate Personal Protective Equipment (PPE), welding equipment usage, and machinery inspections.

A welding safety checklist can also be used when performing site walk throughs that help

record the ventilation systems condition, flammable materials on site, and welding hazards and control.

Start by taking a photo of the work area and check if the operator has undergone proper training, understands instructions, and wears appropriate PPE.

Next, evaluate if equipment is in good working condition and the working area is ventilated and free from flammables and obstructions.

Inspect if procedures were followed during and after work.

End the report by providing recommendations and an overall safety rating.

WELDING SAFETY EQUIPMENT

The welding industry requires wearing appropriate PPE when performing weld. It helps protect welders from the unnecessary incident including skin burns, blindness, or overexposure to fumes and gases.

The following welding safety equipment list can protect welders from welding hazards.

1	Welding helmet	wearing a welding helmet, hood, or mask can protect welders from sparks and radiation emitted by the arc that can quickly damage the skin and eyes
2	Eye protection	this includes goggles or hand-held face shields. This would help prevent eye damage or loss of vision.
3	Respirators	this would protect welders from inhalation of harmful fumes and gases to enter the body.

4	Flame-resistant welding gloves	the working hands are the closest proximity to the arc. Wearing flame-resistant welding gloves can lessen the exposure to intense heat and radiation.
5	Protective suits	this could protect the welder from extreme temperatures, flames, and sparks.
6	Welding-specific work boots	this would protect the top of the foot from flames and sparks.

What are the Safety Requirements for Welding?

The safety requirements for welding involves the welder's training, welding machine, work area, and the actual welding operation. These are based on this welding safety checklist.

Item no.	Descriptive of the learner	Yes	No
1	Are all welding operators trained/certified?		
2	Are all instructions clear and understood by operators?		
3	Are all welding operators familiar with the welding machine?		
4	Are all welding operators aware of the type of materials to be weld?		
5	Do all welding operators wear proper PPE?		
6	Are welding operators aware of all safety and emergency protocols?		
Welding Machine			
7	Are all welding cables, wiring, and electrode holder insulation in good working condition?		
8	Are all welding terminals and joints in good working condition?		
9	Is the size of cable suitable for voltage supply?		

Work Area			
Item No.	Statement	Yes	No
1	Is work area clean and free of flammables and obstructions?		
2	Is work area well ventilated?		
3	Is there a suitable Class ABC fire extinguisher nearby while welding?		
4	Are combustible floors witted or protected?		
5	Are combustible walls shielded or guarded?		
6	Is fire blanket available within the area?		
7	Are warning signs placed within the area?		
8	Is work area restricted to regular employees?		

How to Properly Dress a Wound

Properly dressing a wound is one of the most basic first aid techniques. The process remains unchanged regardless of the size or severity of the injury. Whether it's a child's skinned knee or a gunshot wound, the tenets of wound dressing are identical.

Minor cuts and scratches can be treated at home or on the road. Larger lacerations may also need to be dressed until medical help can be obtained. Cleanliness is key.



Assess Bleeding

Coming in contact with someone else's blood poses risks, such as transmitting certain diseases. If at all possible, protect yourself by following universal precautions and wearing personal protective equipment (e.g., nitrile gloves or a mask).

Then, assess the bleeding. A little bleeding is okay because it helps flush dirt and other contaminants out of the wound, but heavy bleeding is bad.

Call 10177, cell phone call 112 (MTN, Vodacom, CellC and Telkom) if there's:

Bright red or squirting blood

A puncture wound on the head, neck, chest, abdomen, pelvis, or back is more than an inch deep

A deep puncture wound on an arm above the elbow or a leg above the knee

Do what you can to stop the bleeding.

How to Control Bleeding

Clean the Wound

Clean the wound with running water.¹

Wash the skin around the injury with soap. Don't worry if soap gets into the wound, though it is likely to sting and irritate the raw tissue.

Rinse the wound thoroughly to rid it of any dirt and soap.

Use tweezers to remove particles (like broken glass or gravel).

Antibiotic ointment isn't necessary for a wound to heal nicely. However, it can help reduce the pain of raw injuries, such as abrasions.

Cover the Wound

Only cover the wound if it's likely to come in contact with clothing or dirt.

Adhesive bandages are the easiest way to cover most minor lacerations and abrasions.

Cuts less than 2 centimeters long can be held closed with butterfly bandages.

If the edges of a laceration are not easily pulled together, then the wound may need stitches.

Get Medical Help

UNSAFE ACTS & CONDITIONS

Unsafe acts:

An unsafe act is a human error or personal mistake by which there is a chance of accident, injury, property loss, or harm to the environment and the unsafe act can

increase the probability of an accident.

Examples:

1. Failure to wear PPE.
2. Using hands instead of tools and tackles.
3. Lack of knowledge, etc.

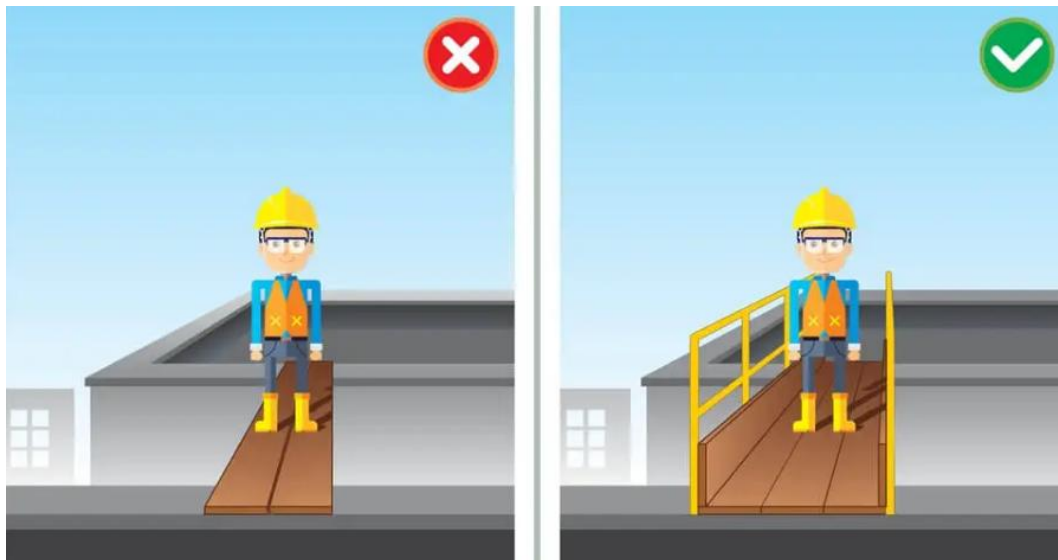


Unsafe condition:

An unsafe condition means the existence of a mechanical, physical, chemical, or environmental condition or situation which may cause a hazard or accident. The unsafe condition must be rectified and repaired.

Examples:

1. Unguarded machine.
2. Poor housekeeping.
3. Presence of flammable gasses.
4. Presence of toxic gasses, etc.



<https://www.youtube.com/watch?v=zMrnt74zVrM>

<https://www.youtube.com/watch?v=VyPd5VULlos>

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Importance of Addressing Unsafe Acts:

- **Prevention of Accidents and Injuries:** Addressing and minimizing unsafe acts can lead to a significant reduction in the number of workplace accidents, injuries, and fatalities.
- **Economic Considerations:** Accidents and injuries can result in significant business costs, including medical expenses, compensation claims, increased insurance premiums, and potential legal actions.
- **Enhanced Productivity:** A safe working environment, free from disruptions caused by accidents, can lead to more consistent and efficient production and operations.
- **Moral and Ethical Responsibility:** Employers have a moral and ethical responsibility to ensure the safety and well-being of their employees. Addressing unsafe acts aligns with this responsibility.
- **Improved Employee Morale:** A safe working environment promotes positive

employee morale. Workers will likely be more motivated and satisfied when they believe their employer prioritizes their safety.

- **Reduction in Downtime:** Accidents can result in downtime, disrupting the regular flow of operations. Addressing unsafe acts can minimize these disruptions.

Workplace Housekeeping Checklist

A workplace housekeeping checklist is a tool used to ensure that the workplace is well organized, hygienic, and safe for all employees. Good housekeeping prevents workplace hazards such as slips, trips, falls, and more.

Regular housekeeping in the workplace also improves productivity, boosts morale, and helps employers avoid potential fines for non-compliance.



Why should we pay attention to housekeeping at work?

Effective housekeeping can help control or eliminate workplace hazards. Poor housekeeping practices frequently contribute to incidents.

Housekeeping is not just cleanliness. It includes keeping work areas neat and orderly, maintaining halls and floors free of slip and trip hazards, and removing of waste materials (e.g., paper, cardboard) and other fire hazards from work areas.

Effective housekeeping is an ongoing operation:

What is the purpose of workplace housekeeping? Poor housekeeping can be a cause of incidents, such as:

tripping over loose objects on floors, stairs, and platforms being hit by falling objects
slipping on greasy, wet,

dirty surfaces striking against projecting, poorly stacked items or misplaced material

cutting, puncturing or tearing the skin of hands or other parts of the body on projecting

nails, wire or steel

strapping

To avoid these hazards, a workplace must “maintain” order throughout a workday.

WHAT ARE SOME BENEFITS OF GOOD HOUSEKEEPING PRACTICES?

Effective housekeeping results in:

- reduced handling to ease the flow of materials
- fewer tripping and slipping incidents in clutter-free and spill-free work areas
- decreased fire hazards
- lower worker exposures to hazardous products (e.g. dust, vapors)
- better control of tools and materials, including inventory and supplies
- more efficient equipment cleanup and maintenance
- better hygienic conditions leading to improved health
- more effective use of space
- reduced property damage by improving preventive maintenance

Equipment: Measuring

Hand tools

Measuring Instruments and Their Functions

A measuring instrument is a tool used to measure the unit value of a quantity. For example, to measure the length of a branch, you can measure it by using a ruler, which means the quantity measured here is the length and the measuring instrument is a ruler.

Steel Ruler



A ruler is a long measuring instrument that is used to measure the length of an object.
Generally, rulers are made of iron, mica or wood.

Measuring Tapes



The Tape measure is an instrument for measuring the length of an object.

Elbow Ruler



The elbow ruler is a measuring instrument used to measure the length as well as to ensure that the right angle is equal to 90°.

Calipers



A caliper is an instrument used to measure the length, thickness, diameter, and depth of an object. It has a measurement accuracy of 0.1 mm.

For example, if we want to measure glass with this tool, we will have the following results:

The outer diameter using the instrument's part no. 1

The inner diameter using the instrument's part no. 2

The glass wall thickness with the instrument's part no. 1

The depth of the glass using the instrument's part no. 3

MICROMETER



A micrometer is also used to measure the diameter and thickness of an object. However, this tool has a higher accuracy than calipers with an accuracy of 0.01 mm.

Accurately measure length, width and thickness in mm

Sliding adjustable square

Other names for the tool include combo square or sliding square.



A combination square is a multi-purpose measuring and marking tool used in metalworking, woodworking, and stonemasonry.

It is composed of a rule and one or more interchangeable heads that can be attached to the rule.

The most common head is the standard head, which is used as a square for marking and testing 90° and 45° angles.

A combination set with the rule (top), a standard head (left), protractor head (centre), and a centre finder head (right)



Illustration showing a centre head, protractor head, and standard head attached to a rule.

Rule

Combination square rules are made of steel and the gradations can be in metric, imperial, or both. The rule typically comes in lengths between 150mm and 600mm or between 4 inches and 24 inches.

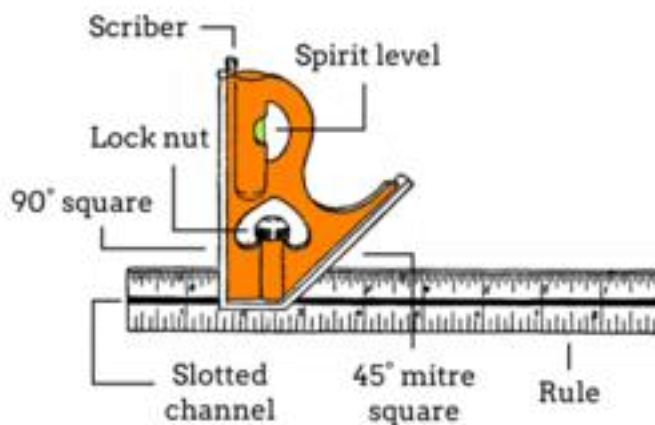
Heads

The heads, are also called anvils, are attached to the rule by sliding the rule into a slot in the side of the head.

The standard head usually incorporates a small spirit level and a small removable

The centre finder head has two faces meeting at 90° , when attached one edge of the rule bisects the two faces at 45° .

The heads are manufactured from either forged steel, cast iron, die-cast aluminium, die-cast zinc, or plastic. Aluminium and zinc heads are cheaper than steel and iron, but less durable and more prone to inaccuracy.



Uses

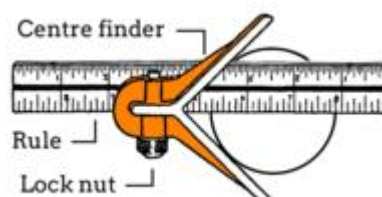
the rule is used in combination with the different heads.

Standard head

The standard head can be used as a: Square, for marking and referencing 90° angles and checking if surfaces are flat and square to one another.

Spirit level, to check if a surface is level or similarly if a surface or edge is plumb (vertical).

Centre finder head



Labeled diagram of the centre finder head with a circle as an example

The centre finder head can be used for:

Marking lines through the centre of circular or square objects, such as dowels. Making multiple marks at different angles can be used to identify the point at the centre of the circle.

Marking lines perpendicular to a curved edge (normal lines).

Bisecting square corners to mark a 45° angle.

Materials

Ferrous & non-ferrous metals

Ferrous metals contain iron and non-ferrous metals have no iron

Ferrous metals are magnetic and non-ferrous metals are not magnetic

Ferrous metals are typically less expensive and non-ferrous metals are typically more expensive

Ferrous metals have lower conductivity and non-ferrous metals can have higher conductivity (depending on the specific non-ferrous metal)

General Properties of Non-Ferrous Metals

Non-ferrous metals have characteristics that make them useful as elemental metals or as alloys. General properties of non-ferrous metals include:

- ◆ They are not magnetic
- ◆ They resist corrosion
- ◆ They are good conductors of electricity and thermal energy
- ◆ They tend to be colorful as compared to ferrous metals making them aesthetically appealing for use in jewelry
- ◆ and other artistic uses of metals
- ◆ They have lower densities

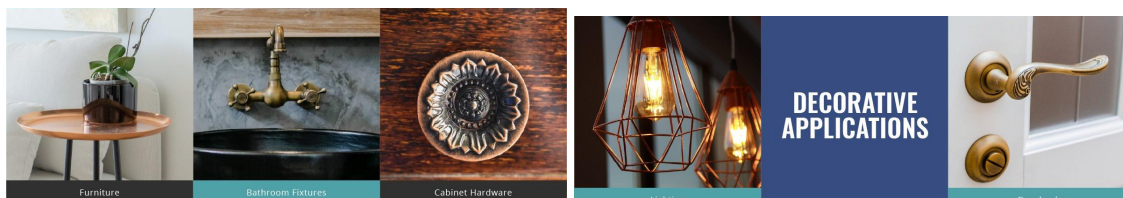
-
- ◆ They have good malleability and ductility, making them good metals for fabrication as they are easily shaped

BRASS

Brass is commonly used in applications where corrosion resistance and low friction are required, such as locks, hinges, gears, bearings, ammunition casings, zippers, plumbing, hose couplings, valves, and electrical plugs and sockets.

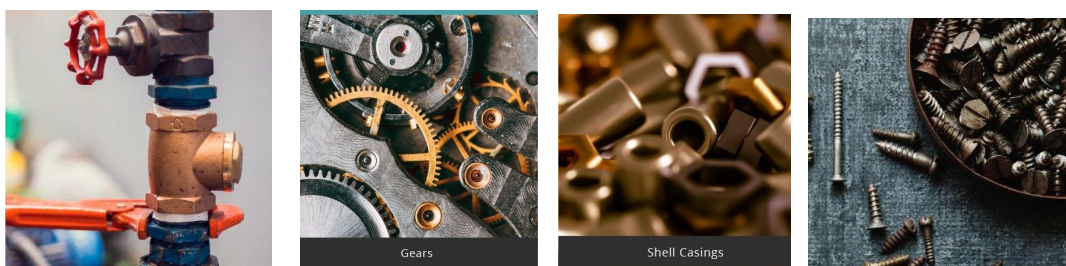
DECORATIVE APPLICATIONS

Residential dishwasher fittings and lamp fittings are commonly made of brass, as they are both visually appealing and bacteria resistant.



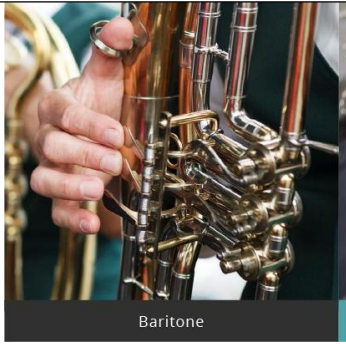
MECHANICAL APPLICATIONS

Tools made of brass are known to have extended life and a reduced need for sharpening.



MUSICAL INSTRUMENTS

If you've ever participated in a concert band, marching band, or perhaps even a symphony, you are likely very aware of the brass all around you. Trumpets, french horns, trombones, baritones, and tubas are some of the most popular brass instruments.



COPPER

Copper has very high thermal conductivity and very high electrical conductivity.

Copper is the most commonly used metal for electrical wiring. It is also used in heating appliances.

Copper has excellent malleability and ductility, which makes it easily spread out into thin wire for electrical wiring.

Copper has good corrosion resistance which makes it beautiful to use in statues and architecture. It can even be used for roofs.

Aluminum

Aluminum is a non-ferrous metal that is light and it has many applications for use in machining and industry. Aluminum is resistant to corrosion, a good conductor of heat, and a good, but not great, conductor of electricity. Aluminum has good ductility and malleability. It is the most malleable among non-ferrous metals. This is what makes it great to use for aluminum soda cans.

Uses of Copper

Kitchen Sink

- Copper is a good choice for kitchen sink because it is generally resistant to corrosion and it has anti-microbial properties.

Table Tops

- As copper is extremely malleable. It can be turned into thin sheets or plates that make a perfect cover for table and counter tops. Copper's anti-microbial property makes

it suitable for this application as well since these are high-touch surfaces.

Copper Jewelry

Door Knobs and Pull Handles

Tools

Musical Instruments

Copper's high electrical conductivity makes it the safest and most efficient material for wire cores. Unlike silver, it doesn't burn up when conducting high voltages.

Pipes

– Copper's antimicrobial property once again proves extremely useful, and this time in filtration applications. Pipes used to convey water or substances requiring zero-level contamination are often made of copper alloys.

wrapping paper

Aluminum foil or aluminum paper is used for wrapping many different items, including food. Aluminum paper is generally safe, easy to use and can be recycled easily. They do not cause contamination or food poisoning.



Aside from these, aluminum has several other applications in everyday life. This includes product packaging, medicine, paint production, furniture, and so on.

Mild steel

Uses for mild steel include structural steel, signs, automobiles, furniture, fencing, and much more.

Steel Frame Buildings – This type of steel is a popular choice for construction frame materials due to the high strength of mild steel beams.

Gates and Fencing – . Low carbon steel is hard to break and can be painted, primed or

galvanised to prevent corrosion and give it a decorative finish.

Pipelines – Mild steel tubes are a popular choice when looking to create steel pipes for various projects.

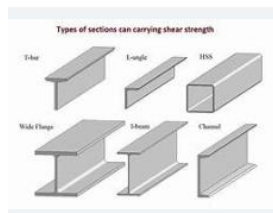
Stainless Steel

Uses and properties

it is used to produce a variety of automotive parts such as exhaust systems, grills, and trims. With advancing technology, stainless steel is being favoured by manufacturers to make structural components.

Stainless steel is preferred in clean and sterile environments as it is simple to clean and does not easily corrode. Stainless is used in the production of a wide range of medical equipment, including surgical and dental instruments.

PROFILES OF THE MATERIALS



Welding process

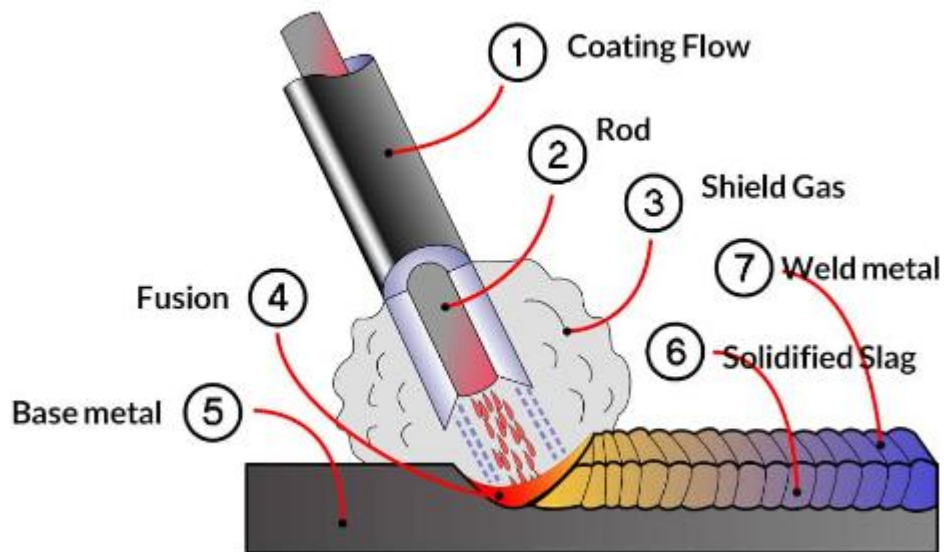
Shielded metal arc welding (smaw)

Shielded metal arc welding (SMAW), also known as stick welding or manual metal arc welding (mma), utilizes a welding arc, which is a sustained electrical discharge through a high-conducting plasma.

How to use shielded metal arc welding (smaw)

To do smaw, follow these steps:

1. Clean the metal surfaces to be welded. Eliminate rust, dirt, and oil residue from the surface to ensure a strong weld..
2. Choose the electrode type based on the metal being welded and the desired weld characteristics.
3. Connect the welding machine's power cable to a suitable power source. Connect the electrode holder to the positive terminal and the ground clamp to the negative terminal of the welding machine.
4. Hold the electrode at a slight angle and strike it gently against the workpiece to initiate the arc. Ensure proper alignment, and hold the welding electrode in a vertical position relative to the area where the materials will be joined. Subsequently, tilt the electrode slightly at an angle ranging from around 5 to 15 degrees.
5. Move the electrode in a steady and controlled manner along the joint, maintaining the correct arc length and angle.
6. Allow the molten metal from the electrode to fill the joint, which will ensure proper penetration and fusion.
7. Ensure even heat distribution and proper fusion by carefully monitoring and adjusting the welding speed. Additionally, adjusting the technique can help control the weld puddle and produce the desired bead size and appearance.
8. Continue welding until the joint is fully traveled along the entire length of the intended weld, then extinguish the arc by carefully removing the electrode from the workpiece while still in control of the welding machine.
9. Remove the slag created by the flux from the weld bead surface and inspect the weld for any defects, such as: Cracks, blow holes, areas of obvious lack of penetration, etc.



What basic equipment is required for smaw?

The basic equipment required for smaw includes:

1. **Welding machine or power source:** Provides the electric current for welding.
2. **Electrode holder:** Holds the smaw electrode and connects it to the welding machine.
3. **Ground clamp:** Completes the electrical circuit between the welding machine and the workpiece.
4. **Smaw electrodes:** Consumable electrodes with flux coatings, selected based on the type of metal being welded.
5. **Personal protective equipment (ppe):** Includes welding helmet, gloves, protective clothing, and safety glasses for operator safety.

What is the right speed for shielded metal arc welding?

The recommended range of travel speed for shielded metal arc welding (smaw) is generally between 75-150 mm/minute. This speed range ensures that enough heat is generated to provide adequate fusion.

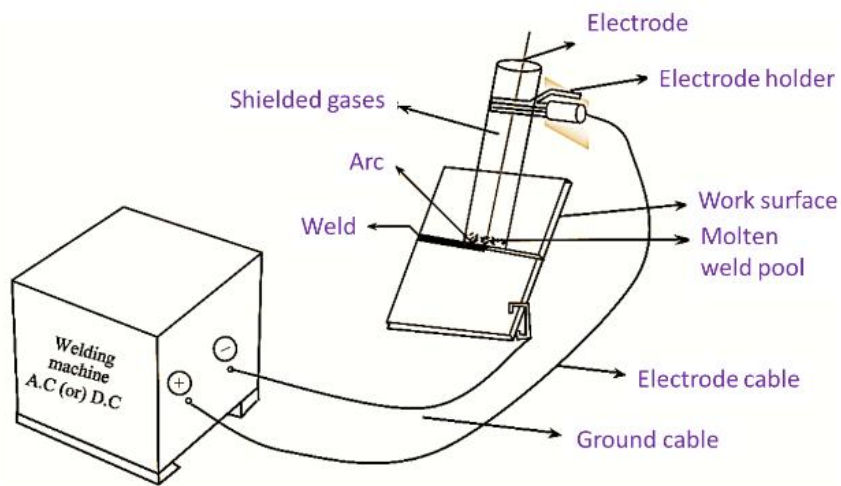


Figure 1: Shielded Metal Arc Welding.

What is the recommended power to shielded metal arc welding?

For most welding purposes, the welding voltage typically ranges from 20 to 80 volts (v), while the current ranges from 30 to 1500 amperes

Different welding processes use different current types, such as direct current (dc), alternating current (ac), or pulsed current. Waveform control/pulse welding, constant current welding, and constant voltage welding are the three types of power sources for welding.

In what positions can smaw be performed?

The five welding positions for shielded metal arc welding (smaw) are flat, horizontal, overhead, vertical-up, and vertical-down.

Each position requires speed and technique adjustments to achieve the desired weld quality.

When working with materials thicker than sheet metal, slowing down the travel speed enhances penetration and fusion. This promotes a stronger and more credible weld joint in vertical-up positions by ensuring that the molten metal has enough time to cool and solidify.

Which metals can be welded with smaw?

The following metals can be welded with shielded metal arc welding (smaw):

1. Stainless steel

2. Tool steel
3. Nickel
4. Cast iron
5. Carbon steel

What material thicknesses can be welded with smaw?

Smaw, or stick welding, is commonly used to weld metal with a thickness of 3.2 mm or thicker.

What industries use shielded metal arc welding (smaw)?

The industries that commonly use smaw include:

1. Construction
2. Infrastructure
3. Shipbuilding
4. Automotive manufacturing and repair
5. Oil and gas
6. Pipeline construction and maintenance
7. Structural steel fabrication
8. Heavy equipment manufacturing and repair

What are the advantages of shielded metal arc welding (smaw)?

The advantages of smaw include:

1. Portability and versatility
2. Suitability for outdoor and remote work environments
3. Cost-effectiveness compared to other welding methods
4. Ability to weld a wide range of metals and alloys
5. Deep penetration capability
6. Ability to weld thick materials
7. Minimal equipment requirements

What are the disadvantages of shielded metal arc welding (smaw)?

The disadvantages of smaw include:

1. Slower welding process compared to some other methods such as gas tungsten

arc welding (gtaw) and gas metal arc welding (gmaw)

2. Limited efficiency due to high heat loss
3. Smaw generates higher levels of welding fumes compared to welding processes such as gmaw and gtaw.
4. Smaw results in the formation of slag during the welding process.
5. Requires frequent electrode changes
6. Limited suitability for thin materials due to potential distortion

What is the difference between smaw and fcaw?

Smaw (shielded metal arc welding) and fcaw (flux-cored arc welding) are both arc welding processes with some key distinctions. While fcaw uses a tubular electrode filled with flux, smaw employs a solid electrode coated with flux. This difference allows fcaw to achieve higher deposition rates compared to smaw.

What is the difference between smaw and gmaw?

Both smaw (shielded metal arc welding) and gmaw (gas metal arc welding) are arc welding processes. However, the electrode and shielding techniques used in each process vary. While gmaw uses a continuous wire electrode and relies on a shielding gas, smaw uses a flux-coated electrode and generates liquid slag to protect the molten weld pool.

Proper storage of welding cables and equipment is crucial for ensuring safety and maintaining the longevity of the tools.

storing welding cables in a figure-eight coil pattern can help prevent unnecessary twisting and tangling of the cables?

Proper care of: Welding helmet

Cleaning Process

Cleaning welding helmet is extremely important,

#1 Focus on the Lens

The best place to start cleaning your welding helmet is the lens. The lens is the dirtiest part of a welding helmet. So, be prepared to combat some dust and particles.

#2 Inspect Everything

Once you're done cleaning the lens, it's important to inspect all the other aspects of the welding helmet. Start from the bottom of the front of your welding helmet and work your way to the top and the back.

#3 Clean the Dust

After inspecting the rest of your welding helmet, simply clean the dust off your entire helmet.

#4 Tighten Screws

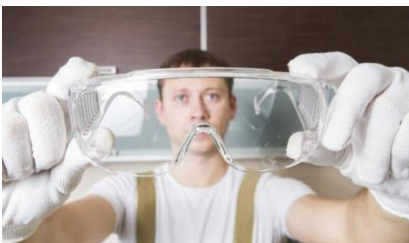
you should examine all of the screws and make sure they're tight enough. If any of the screws are loose, simply take a screwdriver to fix them.

#5 Store Your Helmet Properly

After cleaning your welding helmet, it's important to properly store it.

If you have a battery powered helmet, then make sure the batteries are not in the helmet when you're cleaning it.

Safety Glasses Maintenance



Clean daily.

Cleaning can be as simple as rinsing glasses under cool water to remove debris and then drying them with a soft cloth. Spraying with a can of compressed air can

remove dust and other particles that may have accumulated. Ideally, keep a cleaning cloth specially made for glasses handy.

Consider storage.

The best option for storing glasses is in a hard case or microfiber pouch especially made for safety eyewear. A Microfiber Pouch is a great option too because it doubles as a cleaning cloth for lenses.

An eyewear retainer or neck cord allows glasses to hang from the wearer's neck and keeps them to readily available.

Leather gloves

Cold water is Saddle soap is specifically made to clean dirt and grime off of leather products, including welding gloves. It is important to use a soft-bristled brush to avoid damaging the leather while cleaning. A clean cloth or sponge is needed to wipe away the soap and water after cleaning.

Step 1: Preparing the Gloves

Use a soft-bristled brush to remove any dirt or debris from the gloves. Be sure to brush gently, so as not to damage the leather.

If there are any stubborn stains or marks on the gloves, use a damp cloth to gently rub them away. Avoid using any harsh cleaning agents or chemicals, as these can damage the leather.

Once you have removed any dirt or debris from the gloves, it's time to move on to the cleaning process.

Step 2: Cleaning the Gloves with Soap and Water

Pay special attention to any areas that are particularly dirty or stained.

Rinse the gloves thoroughly with clean water to remove any soap residue.

Gently squeeze out any excess water from the gloves, being careful not to twist or wring

them.

Lay the gloves flat on a clean towel to dry, away from direct heat or sunlight.

It's important not to use too much water when cleaning leather welding gloves, as this can cause the leather to become saturated and lose its shape.

Step 3: Treating the Gloves with Leather Conditioner

Leather conditioner helps to restore the natural oils in the leather, keeping it soft and flexible.

Gently rub the conditioner into the leather, paying particular attention to any areas that are prone to dryness or cracking, such as the fingertips and seams.

Allow the conditioner to soak into the leather for several minutes, then wipe away any excess with a clean, dry cloth.

Step 4: Drying the Gloves

After washing the gloves, the next step is to dry them properly. It is important to dry them completely to avoid any mold or mildew growth. Here are some ways to dry leather welding gloves:

Air Drying

Air drying is the most effective way to dry leather welding gloves. Hang the gloves in a well-ventilated area and let them dry naturally. Do not put them in direct sunlight or near any heat source, as this can cause the leather to crack or shrink.

Towel Drying

Using a Fan

Using a Hair Dryer

If you are in a hurry, you can use a hair dryer to dry the gloves. Remember, it is important to dry the gloves completely before using them again. Failure to do so can result in mold or mildew growth, which can be harmful to your health.

Step 5: Storing the Gloves

After cleaning the leather welding gloves, it is important to store them properly to ensure they remain in good condition. Here are some tips on how to store your gloves:

7. Apply a variety of welding, oxy-fuel cutting and oxy-fuel joining processes.